

BRANDON ELIJAH LE

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Education

University of California, San Diego

M.S. Computer Science

January 2024 – June 2025

University of California, San Diego

B.S. Computer Science

October 2020 – December 2023

Courses: Advanced Data Structures & OO Design, Computer Organization & System Programming, Software Engineering, Design & Analysis of Algorithms, Computer Security, Machine Learning, Recommender Systems, Network Systems

Experience

F5 Networks

Seattle, WA

Software Engineer Intern

June 2024 – September 2024

- Designed and implemented a simplified internal document store table to replace complex internal tables, optimizing performance, memory, and disk usage.
- Developed and maintained services using Golang, gRPC, PostgreSQL, and Open API, improving the efficiency and scalability of the control plane components.
- Contributed to the development and enhancement of the BIG-IP Next control plane, ensuring smooth and secure application functionality for customers and users.
- Participated in weekly sprint planning, daily standup, code reviews, and collaborative problem-solving sessions

UC San Diego Computer Science & Engineering Department

San Diego, CA

Teaching Assistant

Sept 2021 – Present

- Guided hundreds of students, strengthening their foundation in key courses such as Data Structures, Object-Oriented Design, and Intro to Programming, through dedicated mentorship and detailed explanations.
- Facilitated individualized learning experiences in one-on-one tutoring sessions, resulting in improved student comprehension of complex concepts and problem-solving skills.
- Collaborated with faculty, contributing to curriculum development and the continuous improvement of instructional methods, thereby driving the program's reputation for excellence.

Projects

CloudStore

San Diego, CA

Cloud Application - GoLang, SQLite, Unix Scripting

March 2024 – June 2024

- Designed and implemented a network file storage using gRPC, allowing file synchronization.
- Implemented a Consistent Hashing Ring to efficiently distribute and manage data blocks on the cloud.
- Implemented the data consensus RAFT protocol to achieve data consistency and fault tolerance
- Used SQLite3 to maintain metadata and allowed multiple users to service simultaneously without race conditions.

Playlist Plus

San Diego, CA

Music Recommender System - Python, TensorFlow

September 2023 – December 2024

- Designed and developed a music recommender system using the Spotify Million Playlist Dataset (MPD) to generate playlists based on a single seed playlist, using Bayesian Personalized Ranking (BPR)
- Managed and analyzed a subset of the MPD, containing 10,000 playlists and 670,464 songs, addressing challenges in metadata enrichment and dataset scaling.
- Investigated parallel processing, sampling techniques, as well as collaborative filtering and matrix factorization

Uno Strategy Algorithm & Simulator

San Diego, CA

Turn-Based Decision-Making Algorithm - Python, Jupyter Notebook

January 2024 - March 2024

- Implemented an algorithm for Uno that achieved a 33.5% higher win rate compared to random play strategies
- Utilized adversarial search to evaluate future game states, enabling the algorithm to select the best moves
- Conducted one million simulations using a Jupyter notebook to evaluate the algorithm's performance against random opponents, analyzing win rates, game duration, and turn efficiency.

Languages: Golang, Python, Java, C++, C

Technologies: Git, Bash, Docker, Flask, Figma, SQL